

Generalized Bayes: the route back to standard Bayes

The inference engine FedGVI federates is the generalized (Gibbs) posterior (Bissiri et al. 2016; Jiang and Tanner 2008; Jewson et al. 2018; Knoblauch et al. 2022),

Generalized Bayes (part 2)

which trades the likelihood for a loss L and the KL regularizer for a general divergence $\mathcal{D}$:

Generalized Bayes (part 3)

$$q_n^*(s) = \arg \min_{q_n} \left\{ \mathbb{E}_{q_n} \left[\sum_i L(s; o_i) \right] + \frac{1}{\tau} \mathcal{D}(q_n \parallel \pi_0) \right\}. \quad (2)$$

Generalized Bayes (part 4)

with prior π_0 , learning rate τ , and regularizing divergence $\mathcal{D}$. The learning rate is part of the inferential specification, not a cosmetic constant;

Generalized Bayes (part 5)

coarsened-posterior and safe-Bayes work show why calibration of that temperature matters under misspecification (Miller and Dunson 2018; Grünwald 2012), where ordinary Bayes concentrates around a KL pseudo-truth rather than literal truth when the model family is wrong (Kleijn and Vaart 2012).

Generalized Bayes (part 6)

We name the object eq. 2 defines.

Generalized Bayes (part 7)

Definition (Generalized-(Gibbs)-Bayes posterior)

Given loss L , prior π_0 , learning rate $\tau > 0$, and divergence $\mathcal{D}$, the generalized-Bayes posterior q_n^* minimizes (2). When $\mathcal{D} = \text{KL}$, its exact categorical minimizer is the tempered softmax in (3).

Generalized Bayes (part 8)

$$q_n^*(s) \propto \pi_0(s) \exp(-\tau \sum_i L(s; o_i)), \quad (3)$$

Generalized Bayes (part 9)

The implementation surface is the `generalized_posterior` function in the `generalized_bayes` module.

Generalized Bayes (part 10)

The tempered softmax of eq. 3, stated in the definition above, is not an approximation: it is the exact closed-form minimizer of eq. 2 when the regularizer is the KL divergence, because the categorical support is finite and the objective is strictly convex in q .

Generalized Bayes (part 11)

The recovery to standard Bayes follows by choosing the loss.

Generalized Bayes (part 12)

With $L = \text{NLL}$, $\text{NLL}(p, o) = -\log p(o)$, the exponential in that tempered softmax becomes a product of likelihoods and the minimizer is *exactly* standard Bayes; eq. 11 in sec. 5.8 states that corner,

Generalized Bayes (part 13)

and Corollary 6 there pins it to the closed-form prior-times-likelihood product.

Generalized Bayes (part 14)

The largest observed discrepancy between `generalized_posterior` in this regime and the analytic Bayes posterior is $5.55\text{e-}17$, reported in sec. 7.1 — exact to machine precision (a maximum deviation of about one ULP), not merely close.

Generalized Bayes (part 15)

FedGVI computes each client update against a *cavity* rather than the full posterior, so a contributing agent does not double-count its own previous message. We name that operation.

Generalized Bayes (part 16)

Definition (Cavity / PVI factor update)

The cavity is the normalized removal of agent n 's site factor from the colony posterior in natural-parameter space, as in (4). PVI replaces the site factor and recombines it with that cavity according to (5).

Generalized Bayes (part 17)

$$\begin{aligned} q_{-n}(s) &= \frac{q(s)/t_n(s)}{\sum_{s'} q(s')/t_n(s')} \\ &= \text{softmax}(\log q(s) - \log t_n(s)), \end{aligned} \tag{4}$$

Generalized Bayes (part 18)

The final expression makes normalization explicit. Taking a cavity and re-multiplying the original site factor restores the global posterior:

Generalized Bayes (part 19)

$$q(s) = \frac{q_{-n}(s)t_n(s)}{\sum_{s'} q_{-n}(s')t_n(s')}, \quad (5)$$

Generalized Bayes (part 20)

This is the recombination identity satisfied by `generalized_bayes.cavity` and `generalized_bayes.update_factor`.

The numbered recombination identity is eq. 5.

Generalized Bayes (part 21)

The cavity of eq. 4 is the discrete analogue of the expectation- propagation / partitioned-VI cavity used outside active inference (Ashman et al. 2022; Bui et al. 2018),

Generalized Bayes (part 22)

imported here so that the per-agent generalized-Bayes update of eq. 2 is computed against the colony belief with the agent's own contribution removed — exactly the sensory-attenuation discipline the belief-sharing round of sec. 5.9 requires.

Generalized Bayes (part 23)

What remains unspecified in eq. 2 are its two ingredients — the divergence $\mathcal{D}$ and the loss L — whose robust members and standard-Bayes limits sec. 5.6 develops next;

Generalized Bayes (part 24)

the aggregation identity (sec. 5.8) then federates the resulting per-agent posteriors.

Generalized Bayes (part 25)

The authoritative notation supplement makes the same normalization and recombination contract explicit in eq. 36 and eq. 37; those equations govern the symbols used by the implementation and all later supplements.

Conjugate likelihood learning for the shared model

Active-inference agents learn the parameters of their generative model, not just plan with them (Smith et al. 2022; Friston et al. 2024).

Conjugate likelihood... (part 2)

The likelihood matrix A carries a Dirichlet prior with concentration a over each column, updated conjugately by accumulating observation-state co-occurrence counts (eq. 18),

Conjugate likelihood... (part 3)

giving the column-normalized expected likelihood. The update of eq. 18 is driven by the expected sufficient statistics under the data-generating model,

so as the concentrations accumulate $\mathbb{E}[A]$ converges to the true likelihood.

Conjugate likelihood... (part 4)

Convergence is measured by the per-column KL divergence summed over hidden states, which decreases monotonically toward the standard-Bayes fixed point; sec. 7.3 reports the learning curve, where the KL falls from 3.4231 to 0.0027 across 24 count batches.

Conjugate likelihood... (part 5)

A forgetting hyperprior optionally decays the running mass toward an asymptote so the agent does not become infinitely confident; with the hyperprior disabled the classical unbounded accumulation of eq. 18 is recovered.

Conjugate likelihood... (part 6)

The implementation is `learn_likelihood` in the `dirichlet_learning` module.

Bayesian model reduction for structure comparison

Structure learning in the active-inference frame proceeds by Bayesian model reduction (BMR): given a full model with Dirichlet posterior post under prior prior,

Bayesian model reduction... (part 2)

the change in negative variational free energy from swapping in a *reduced* prior — for example one that prunes a redundant column toward zero —

Bayesian model reduction... (part 3)

is available in closed form without re-running inference (Friston and Penny 2011; Smith et al. 2022).

Bayesian model reduction... (part 4)

Because the likelihood is shared, the reduced posterior is $\text{post} + \text{reduced_prior} - \text{prior}$, and the free-energy difference is a difference of log multivariate Beta functions (eq. 20), where $\ln B(a) = \sum_k \ln \Gamma(a_k) - \ln \Gamma(\sum_k a_k)$ is the log Dirichlet normalizer.

Bayesian model reduction... (part 5)

A positive ΔF in eq. 20 means the reduced model has more evidence — the pruned structure was redundant and should be adopted; a negative ΔF means the reduction destroyed something the data support. When the reduced prior equals the prior the score is identically zero,

Bayesian model reduction... (part 6)

the no-reduction fixed point.

sec. 7.4 reports $\Delta F = 3.68$ for a redundant reduction (accepted) against $\Delta F = -27.67$ for a supported one (rejected). The implementation is `bayesian_model_reduction.reduce`.